Courses Managem ent System | **The Code**

*CoursesManagementSystem.java

```java
package ProjectOOP;

import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;

// Class representing a course
class Course {
    private String title;
    private String instructor;
    private String area;

    // Constructor
    public Course(String title, String instructor, String area) {
        this.title = title;
        this.instructor = instructor;
        this.area = area;
    }

    // Getters and setters
    public String getTitle() {
        return title;
    }

    public void setTitle(String title) {
        this.title = title;
    }

    public String getInstructor() {
        return instructor;
    }

    public void setInstructor(String instructor) {
        this.instructor = instructor;
    }

    public String getArea() {
        return area;
    }
    public void setArea(String area) {
        this.area = area;
    }

    public String toString() {
        return "Title: " + title + ", Instructor: " + instructor + ", Area: " + area;
    }
}

// Class representing the supervisor, managing courses
class Supervisor {
    private List<Course> courses = new ArrayList<>();
    // Method to add a new course
    public void addCourse(Course course) {
        courses.add(course);
        System.out.println("Course added successfully!");
    }
    // Method to display all courses
    public void displayCourses() {
        if (courses.isEmpty()) {
            System.out.println("No courses available.");
        } else {
            courses.forEach(System.out::println);
        }
    }
    // Method to search for courses by title or instructor
    public void searchCourses(String keyword) {
        boolean found = false;
        for (Course course : courses) {
            if (course.getTitle().equalsIgnoreCase(keyword) || course.getInstructor().equalsIgnoreCase(keyword)) {
                System.out.println(course);
                found = true;
            }
        }
        if (!found) {
            System.out.println("No matching courses found.");
        }
    }

    // Method to display courses by area
    public void displayCoursesByArea(String area) {
        boolean found = false;
        for (Course course : courses) {
            if (course.getArea().equalsIgnoreCase(area)) {
                System.out.println(course);
                found = true;
            }
        }
        if (!found) {
            System.out.println("No courses found for the area: " + area);
        }
    }
}

// Main class with CLI for user interaction
public class CoursesManagementSystem {
    public static void main(String[] args) {
        Supervisor supervisor = new Supervisor();
        Scanner scanner = new Scanner(System.in);

        // Adding sample data
        supervisor.addCourse(new Course("Introduction to Islam", "Thuraiah Alqasem", "Islam"));
        supervisor.addCourse(new Course("Linear Algebra", "Nevine Gunaime ", "Mathematics"));
        supervisor.addCourse(new Course("Object Oriented Programming", "Eman Alyahya", "Data Science"));
        supervisor.addCourse(new Course("Technical Writing", "Rasha Alshaye", "English"));


        while (true) {
            System.out.println("\nWelcome to the Courses Management System");
            System.out.println("1. Supervisor Menu");
            System.out.println("2. Student Menu");
            System.out.println("3. Exit");
            System.out.print("Choose an option: ");
            int choice = scanner.nextInt();
            scanner.nextLine(); // Consume newline

```

Courses Management System | **The Code**

*CoursesManagementSystem.java

```java
            if (choice == 1) {
                System.out.println("\nSupervisor Menu:");
                System.out.println("1. Add Course");
                System.out.println("2. Display All Courses");
                System.out.println("3. Search Course by Title or Instructor");
                System.out.println("4. Display Courses by Area");
                System.out.print("Choose an option: ");
                int supervisorChoice = scanner.nextInt();
                scanner.nextLine(); // Consume newline

                switch (supervisorChoice) {
                    case 1:
                        System.out.print("Enter course title: ");
                        String title = scanner.nextLine();
                        System.out.print("Enter instructor name: ");
                        String instructor = scanner.nextLine();
                        System.out.print("Enter course area: ");
                        String area = scanner.nextLine();
                        supervisor.addCourse(new Course(title, instructor, area));
                        break;
                    case 2:
                        System.out.println("\nAll Courses:");
                        supervisor.displayCourses();
                        break;
                    case 3:
                        System.out.print("Enter title or instructor name to search: ");
                        String keyword = scanner.nextLine();
                        supervisor.searchCourses(keyword);
                        break;
                    case 4:
                        System.out.print("Enter course area to display: ");
                        String searchArea = scanner.nextLine();
                        supervisor.displayCoursesByArea(searchArea);
                        break;
                    default:
                        System.out.println("Invalid option. Please try again.");
                }
            } else if (choice == 2) {
                System.out.println("\nStudent Menu:");
                System.out.print("Enter title or instructor name to search: ");
                String studentKeyword = scanner.nextLine();
                supervisor.searchCourses(studentKeyword);
            } else if (choice == 3) {
                System.out.println("Exiting the system. Goodbye!");
                break;
            } else {
                System.out.println("Invalid option. Please try again.");
            }
        }
    }
}
```

Courses Management System |**The Output**

Before Testing

```
Problems  Javadoc  Declaration  Console ×
CoursesManagementSystem [Java Application] C:\Users\mayar\Downloads\eclipse-jav
Course added successfully!
Course added successfully!
Course added successfully!
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option:
```

Courses Management System |**The Output(**The testing)

```
Course added successfully!
Course added successfully!
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 1

Supervisor Menu:
1. Add Course
2. Display All Courses
3. Search Course by Title or Instructor
4. Display Courses by Area
Choose an option: 1
Enter course title: Linear Algebra
Enter instructor name: Nevine Gunaime
Enter course area: Mathematics
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 2

Student Menu:
Enter title or instructor name to search: Eman Alyahya
Title: Object Oriented Programming, Instructor: Eman Alyahya, Area: Data Science

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 3
Exiting the system. Goodbye!
```

```
Problems  Javadoc  Declaration  Console
<terminated> CoursesManagementSystem [Java Application] C:\Users\mayar\Downloads\eclipse-java-2024-06-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x
Course added successfully!
Course added successfully!
Course added successfully!
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 1

Supervisor Menu:
1. Add Course
2. Display All Courses
3. Search Course by Title or Instructor
4. Display Courses by Area
Choose an option: 1
Enter course title: Introduction to Islam
Enter instructor name: Thuraiah Alqasem
Enter course area: Islam 101
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 3
Exiting the system. Goodbye!
```

```
Course added successfully!
Course added successfully!
Course added successfully!
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 3
Exiting the system. Goodbye!
```

Courses Management System |**The Output(**The testing)

```
Course added successfully!
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 2

Student Menu:
Enter title or instructor name to search: Technical Writing
Title: Technical Writing, Instructor: Rasha Alshaye, Area: English

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 1

Supervisor Menu:
1. Add Course
2. Display All Courses
3. Search Course by Title or Instructor
4. Display Courses by Area
Choose an option: 2

All Courses:
Title: Introduction to Islam, Instructor: Thuraiah Alqasem, Area: Islam
Title: Linear Algebra, Instructor: Nevine Gunaime , Area: Mathematics
Title: Object Oriented Programming, Instructor: Eman Alyahya, Area: Data Science
Title: Technical Writing, Instructor: Rasha Alshaye, Area: English

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 3
Exiting the system. Goodbye!
```

```
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 1

Supervisor Menu:
1. Add Course
2. Display All Courses
3. Search Course by Title or Instructor
4. Display Courses by Area
Choose an option: 3
Enter title or instructor name to search: Rasha Alshaye
Title: Technical Writing, Instructor: Rasha Alshaye, Area: English

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 1

Supervisor Menu:
1. Add Course
2. Display All Courses
3. Search Course by Title or Instructor
4. Display Courses by Area
Choose an option: 4
Enter course area to display: English
Title: Technical Writing, Instructor: Rasha Alshaye, Area: English

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 3
Exiting the system. Goodbye!
```

```
Course added successfully!
Course added successfully!
Course added successfully!
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 2

Student Menu:
Enter title or instructor name to search: Thuraiah Alqasem
Title: Introduction to Islam, Instructor: Thuraiah Alqasem, Area: Islam

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 3
Exiting the system. Goodbye!
```

Courses Management System |**The Output(**Error Handling)

```
Problems  Javadoc  Declaration  Console ×
CoursesManagementSystem [Java Application] C:\Users\mayar\Downloads\eclipse-java-2024-06-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.3.v2
Course added successfully!
Course added successfully!
Course added successfully!
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 2

Student Menu:
Enter title or instructor name to search: Asma Alghamdi
No matching courses found.

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 2

Student Menu:
Enter title or instructor name to search: Introduction to Data Science
No matching courses found.

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 4
Invalid option. Please try again.

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option:
```

```
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option:
1

Supervisor Menu:
1. Add Course
2. Display All Courses
3. Search Course by Title or Instructor
4. Display Courses by Area
Choose an option: 3
Enter title or instructor name to search: Mohammed Algarni
No matching courses found.

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 2

Student Menu:
Enter title or instructor name to search: Amera Alharbi
No matching courses found.

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 7
Invalid option. Please try again.

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 3
Exiting the system. Goodbye!
```

Courses Management System |**The Output(**Error Handling)

```
Course added successfully!
Course added successfully!
Course added successfully!
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 1

Supervisor Menu:
1. Add Course
2. Display All Courses
3. Search Course by Title or Instructor
4. Display Courses by Area
Choose an option: 4
Enter course area to display: Computer science
No courses found for the area: Computer science

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 0
Invalid option. Please try again.

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 3
Exiting the system. Goodbye!
```

```
Course added successfully!
Course added successfully!
Course added successfully!
Course added successfully!

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 2

Student Menu:
Enter title or instructor name to search: Finance Management
No matching courses found.

Welcome to the Courses Management System
1. Supervisor Menu
2. Student Menu
3. Exit
Choose an option: 3
Exiting the system. Goodbye!
```